

Adeel Ahmad

Email : adeel.ahmad.3a@gmail.com

Webpage: adl1995.github.io

LinkedIn: [adeel-ahmad3a](#)

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Master of Science in Computer Science; GPA: 3.6

Dec 2024

Coursework: Machine Learning, Computer Vision, Computational Photography.

FAST National University

Islamabad, Pakistan

Bachelor of Science in Computer Science

Jun 2018

WORK EXPERIENCE

GÉANT Association | Software Engineer

Cambridge, UK (Oct 2023 – Present)

- Led the migration of 50+ services to our new IdP, transitioning SAML integrations to modern OIDC protocol.
- Designed **authentication** middlewares for API gateways using OAuth2 and configured Puppet services.
- Built FastAPI application to execute Linux commands on routers via SSH for network troubleshooting.
- **Administered Linux systems**, managing reverse proxies and rate-limiting (HAProxy, Nginx) [↗](#).

CERN | Software Engineer

Geneva, Switzerland

Identity & Access Management Team

Sep 2020 – Aug 2023

- Designed the **CERN 2FA system** in Keycloak and executed successful rollout to 20k+ users.
- Built CI/CD pipelines to reduce deployment time by 4 mins and automated server configuration using Puppet.
- Developed a **compromised password check** to secure \$100M LHC machines; resolved 1k+ cases.
- Set up **Grafana** monitoring with Prometheus for system downtime alerts.

Computer Security Team

Sep 2018 – Oct 2019

- Developed a Python incidence response system for copyright infringements. Automated email alerts to users.
- Built an **anomalous login detection** system in Go, alerted users of logins from outside Switzerland/France.

Google Summer of Code | Software Engineer

Remote

Boost C++ Organization

May – Aug 2018

- Devised a **C++ algorithm for computing distance** between antipodal points on the Earth. [Blog post](#) [↗](#).
- Improved accuracy by 5% and reduced execution time by 10% over existing approach.

Open Astronomy Organization

May – Aug 2017

- Developed a Python library to visualize **astronomical images**, reducing fetch latency by 75%. [Blog post](#) [↗](#).

OPEN-SOURCE PROJECTS

ObjectSense: iOS app for real-time on-device object recognition using the YOLO model.

FastRAG: End-to-end RAG pipeline using LangChain and HuggingFace embeddings, running on Vertex AI.

Trip Planner: Programmatic trip planning tool using Google My Maps. ([Hacker News Post](#) [↗](#))

CONFERENCES & TALKS

Conference on Computing in High Energy Physics (Norfolk, USA)

May 2023

“Improving computer security in HEP with multiple factor authentication: experience and reflections” [↗](#)

High Energy Physics Information Exchange (Umeå, Sweden)

Nov 2022

“Enforcing 2FA at CERN: A Technical Report on Our Experiences with User Migration” [↗](#)

SKILLS

Programming: Python, C/C++, PHP, Go, Java, Bash, SQL, JavaScript

Technologies: Docker, Keycloak, PyTorch, Kubernetes, OpenCV, SwiftUI, React, AWS, Puppet

TECHNICAL TRAINING

Kubernetes for Developers

Mar 2023

A hands-on course explaining the Kubernetes architecture and how the deployment configuration works.

Red Hat Linux System Administration

Nov 2020

Covered filesystems (BTRFS, VFS), drive encryption, file ACLs, PAM, networking tools, firewalls, and systemd.